

PETER KIMELI

Senior DevSecOps & AI Platform Engineer

kimelipeter20@gmail.com | +254 708 995 320 | Nairobi, Kenya

PROFESSIONAL SUMMARY

Senior DevSecOps and AI Platform Engineer with 8+ years of experience designing, deploying, and securing complex cloud-native infrastructure at scale, now leading in-house AI and automation initiatives on real production data. Proven expertise in Kubernetes, Terraform, observability (Prometheus, Grafana, Loki, Thanos, OpenTelemetry), and CI/CD pipeline security at enterprise organizations including Safaricom PLC and Cashia Limited. Builds explainable, from-scratch AI/ML systems, Retrieval-Augmented Generation (RAG) agents, Model Context Protocol (MCP) orchestration, and AIOps anomaly detection rather than relying on black-box third-party APIs. Holds CKS, CKA, CKAD, and HashiCorp Terraform Associate certifications and is pursuing an MSc in Artificial Intelligence. Experienced incident commander for P1/P2 incidents, with a track record of driving measurable improvements in system reliability, security posture, and engineering team productivity.

CORE COMPETENCIES

Cloud Platforms	AWS (EKS, EC2, S3, IAM, Lambda, API Gateway), GCP, Azure
Containers & Orchestration	Kubernetes (CKS/CKA/CKAD certified), Docker, Helm, Tekton
Infrastructure as Code	Terraform (HashiCorp Associate certified), Pulumi, Crossplane, Ansible
GitOps & CI/CD	ArgoCD, Flux, GitLab CI/CD, Jenkins, GitHub Actions
Observability	Prometheus, Thanos, Grafana, Loki, Tempo, OpenTelemetry, Jaeger, Alertmanager, Dynatrace
AI/ML & AIOps	RAG, Model Context Protocol (MCP), LLMs (Phi-3, Flan-T5, Mistral, Ollama), sentence-transformers, FAISS, pgvector, knowledge graphs, contextual bandits, gradient boosting, Facebook Prophet, Isolation Forest, AWS SageMaker, Z-Score techniques
Security (DevSecOps)	SonarQube, Veracode, Trivy, Kyverno, JFrog Artifactory, OWASP, Pact Broker
Languages & Scripting	Python, Go, Bash, Java, Kotlin, Node.js
Messaging & Data	Apache Kafka, RabbitMQ, MongoDB, PostgreSQL
QA & Testing	Robot Framework, K6, Artillery, Kobiton, Keptn, Selenium, ReportPortal

PROFESSIONAL EXPERIENCE

AI & Automations Engineer Lead Aug 2026 – Present

Cashia Limited | Nairobi, Kenya

- Architected orchestration pipelines connecting multiple AI models, internal APIs, and data sources, adopting Model Context Protocol (MCP) standards to standardise how AI agents interact with tools and services across Cashia's technology ecosystem.
- Designed an Agentic Payment MCP server fronting the Cashia payments gateway with a mandate/consent layer and policy engine, and prototyped three ML components: gradient-boosted transaction risk scoring, a contextual-bandit model for payment-channel routing, and TF-IDF intent parsing.
- Built a Retrieval-Augmented Generation (RAG) AI agent that answers customer queries directly inside the Cashia mobile and web app, retrieving relevant resolved cases to generate accurate, grounded responses in plain language — reducing dependence on human agents while also assisting the CX team; pulled ~18,600 support records and indexed ~2,200 across 8 categories using sentence-transformers embeddings and a FAISS vector store.
- Designed multimodal AI pipelines capable of processing text, images, audio, and documents to support diverse business use cases across the organisation.
- Developed a knowledge graph layer integrated with RAG pipelines to enhance contextual reasoning over interconnected organisational data.
- Applied AIOps to observability, building time-series anomaly detection with Facebook Prophet to forecast metric baselines and automatically flag deviations across the platform surfacing incidents earlier than static thresholds.

- Continue to lead monitoring and observability across the Cashia platform, owning the full observability stack and integrating it with AI-driven anomaly detection to strengthen reliability and incident response.
- Deploying on-premise / self-hosted inference for data-sensitive workflows serving open-source LLMs on Kubernetes and migrating vector search into cluster Postgres (pgvector) so customer and transaction data never leaves approved, compliant infrastructure.
- Implementing AI safety and guardrail frameworks around production endpoints — output validation, prompt-injection defence, and fail-closed webhook handling drawing on a DevSecOps background to secure AI systems end to end.
- Built and maintain a production Python/FastAPI integration syncing support tickets between Zendesk and Plane, with agent-attributed uploads, scoped project access, and secret-based endpoint authentication.

Senior DevSecOps and Cloud Engineer Jun 2025 – Aug 2026

Cashia Limited | Nairobi, Kenya

- Owned and led the full platform engineering and DevSecOps function, spanning cloud infrastructure, CI/CD pipelines, observability, security, and developer tooling across all product lines.
- Architected and provisioned all AWS infrastructure (EKS, VPC, RDS, ElastiCache, S3, IAM, Route53, ALB, ACM, SQS) using Terraform, ensuring consistent, auditable, and zero-drift environments across dev, staging, and production.
- Managed Cloudflare DNS, WAF rules, and CDN configuration using Terraform, enabling version-controlled DNS changes and improving security posture at the edge.
- Deployed and managed production Kubernetes workloads on EKS, including cluster upgrades, node group management, autoscaling, and namespace-level RBAC.
- Implemented GitOps workflows using ArgoCD for all application deployments, enforcing declarative configuration management and enabling rapid, auditable rollbacks.
- Designed and implemented CI/CD pipelines for Android and iOS mobile applications using GitHub Actions with self-hosted Mac Runners, automating build, test, signing, and release to the Play Store and App Store.
- Built and maintained CI/CD pipelines for backend microservices using GitHub Actions and ArgoCD, covering linting, unit tests, container builds, image scanning, and progressive deployments.
- Integrated SonarQube, Trivy, and JFrog Artifactory into CI/CD pipelines, embedding SAST, container scanning, and artifact management as mandatory quality gates across all services.
- Led the Monitoring and Observability function, establishing standardized Golden Signals across all microservices using Prometheus, Thanos, Grafana, Loki, Tempo, and OpenTelemetry.
- Set up and implemented Real User Monitoring (RUM) for web applications using Grafana Alloy and Grafana Faro, gaining visibility into client-side performance and user experience.
- Implemented anomaly alerting in Prometheus using Z-Score recording rules integrated with Istio, improving detection of irregular traffic patterns and service reliability across microservices.
- Acted as the primary Incident Commander during P1/P2 incidents, orchestrating technical resolution teams and ensuring clear, timely communication to senior management and business stakeholders.
- Developed and delivered detailed Post-Incident Review (PIR) reports, focusing on root cause analysis (RCA) and preventative action plans to drive sustained service stability.
- Integrated alerts and on-call workflows with Slack, Alertmanager, and Grafana OnCall, ensuring the right teams are notified with actionable context.
- Set up and integrated ReportPortal for QA testing workflows, enabling centralized reporting, real-time test result analytics, and improved collaboration between QA and DevOps teams.
- Enhanced QA observability by extending test pipelines with Robot Framework automation, reducing manual testing effort and improving release confidence.
- Set up and integrated Pact Broker for contract testing, ensuring API compatibility and reliability across distributed services.
- Collaborated closely with backend engineers, mobile developers, QA, and product teams to define infrastructure requirements, review architecture proposals, and unblock delivery across all squads.
- Developed and enforced DevSecOps best practices, security compliance standards, and infrastructure governance policies across the engineering organization.
- Deployed and managed a self-hosted Headscale VPN server using GitOps (ArgoCD), enabling secure zero-trust network access across environments; administered user and device access policies using Tailscale clients.

Senior DevSecOps Engineer Aug 2021 – Jul 2025

Safaricom PLC | Nairobi, Kenya

- Implemented a comprehensive observability stack by integrating OpenTelemetry, Thanos, OpenSearch, Tempo, and Grafana for real-time monitoring and logging across distributed systems.
- Utilized Thanos for long-term storage of metrics and Grafana for advanced visualization and analysis of system performance.
- Integrated PagerDuty, Alertmanager, and OpenTelemetry to create systems for anomaly detection and alerting, significantly enhancing monitoring and incident response.
- Pioneered anomaly detection in Prometheus time-series data using Z-Score techniques integrated with Istio, significantly improving the reliability of microservices by flagging irregular traffic patterns.
- Designed and implemented a secure, scalable, and highly available cloud infrastructure across AWS (EKS, EC2, S3, IAM).
- Provisioned and managed cloud infrastructure using Terraform, ensuring consistency, repeatability, and security.
- Collaborated with the DevSecOps team in the design and implementation of robust CI/CD pipelines using GitLab and Jenkins.
- Implemented GitOps-based continuous delivery using Flux, automating deployments and ensuring that infrastructure and application configuration were version-controlled and synchronized.
- Leveraged Tekton for building and deploying cloud-native applications, integrating security controls and scans directly into Kubernetes-native CI/CD pipelines.
- Integrated static code analysis (SonarQube) and security scanning (Veracode) into CI/CD pipelines to identify and remediate vulnerabilities early in the development lifecycle.
- Designed and implemented secure architecture for cloud applications, adhering to industry best practices and regulatory requirements.
- Managed the lifecycle of the SonarQube enterprise license, including utilization review from SonarSource, ensuring continuous service availability and audit readiness.
- Participated in service capacity planning by analyzing Prometheus and Thanos historical metrics to forecast future resource needs for critical services, ensuring zero performance degradation during peak load events.
- Conducted regular security assessments and vulnerability scans on infrastructure and applications, ensuring compliance with industry standards and regulations.
- Developed internal tools using Python and Bash for automating data processing workflows and internal application maintenance.
- Implemented the Robot Framework for QA pipelines, automating test cases to ensure high-quality software releases.
- Set up and configured Kobiton for quality assurance mobile testing, enabling automated and manual testing on real devices.
- Mentored junior team members and provided technical guidance on best practices in DevSecOps methodologies.
- During this tenure, delivered two external consultancy engagements via Finsense Africa: KCB Group PLC (Sep 2024–Apr 2025) and Diamond Trust Bank (Jul–Dec 2023) — detailed below.

Senior DevSecOps Consultant (via Finsense Africa) Sep 2024 – Apr 2025

KCB Group PLC | Nairobi, Kenya

- Led the implementation and optimization of enterprise DevSecOps practices, improving software delivery speed, quality, and operational stability across the SDLC.
- Designed and implemented end-to-end CI/CD pipelines in Azure DevOps, automating build, test, release, and rollback processes.
- Provisioned and managed DevOps tooling on Azure Kubernetes Service (AKS) using GitOps (Helm and ArgoCD) for consistency and scalability.
- Deployed and integrated key DevSecOps tools including SonarQube, Nexus, Harbor, Pact Broker, and ArgoCD in production-grade environments.
- Reduced deployment time from hours to minutes through full pipeline automation across commit, acceptance, and release stages.
- Implemented multi-stage CI/CD pipelines (commit, acceptance, release, rollback) with automated quality gates, approvals, and rollback strategies.

- Enabled on-demand deployments to UAT and Production environments with integrated testing and governance controls.
- Implemented enterprise-grade contract testing using Pact Broker, improving service integration reliability across teams.
- Automated functional testing and DAST security testing within CI/CD pipelines, enabling continuous validation.
- Established centralized test execution and reporting, improving traceability to code versions.
- Integrated SAST, SCA, and DAST security scanning into CI/CD pipelines for early vulnerability detection and remediation.
- Implemented DevSecOps governance controls to ensure compliance with enterprise security, audit, and regulatory requirements.
- Enabled continuous security testing and automated compliance checks across environments.
- Strengthened security posture through centralized vulnerability management and reporting mechanisms.
- Enhanced monitoring, logging, and alerting capabilities, enabling proactive incident detection and response.
- Improved system reliability by reducing change failure rates and Mean Time to Recovery (MTTR).
- Established DevOps metrics tracking (deployment frequency, cycle time, quality metrics) to support data-driven decision-making.
- Improved environment consistency, reducing configuration drift and deployment failures.
- Standardized engineering practices and DevOps workflows across development, QA, security, and operations teams.
- Improved cross-team collaboration through shared tooling, automation, and visibility.
- Accelerated time-to-market and improved product quality through automation and early defect detection.
- Reduced manual effort and operational overhead through CI/CD and test automation.

Senior DevSecOps and Cloud Engineer (via Finsense Africa) Jul 2023 – Dec 2023

Diamond Trust Bank (DTB) | Nairobi, Kenya

- Led the implementation of DevSecOps practices and CI/CD pipeline automation, improving software delivery speed and operational stability across the SDLC.
- Designed and implemented end-to-end CI/CD pipelines in Azure DevOps, automating build, test, release, and rollback processes.
- Provisioned and managed DevOps tooling on Azure Kubernetes Service (AKS) using GitOps (Helm and ArgoCD) for consistency and scalability.
- Deployed and integrated key DevSecOps tools including SonarQube, Nexus, Harbor, and ArgoCD in production-grade environments.
- Designed, implemented, and integrated a Robot Framework QA pipeline, enabling automated functional test execution and centralized reporting within CI/CD workflows.
- Established centralized test execution and reporting, improving traceability between test results and code versions.
- Implemented multi-stage CI/CD pipelines (commit, acceptance, release, rollback) with automated quality gates, approvals, and rollback strategies.
- Enabled on-demand deployments to UAT and Production environments with integrated testing and governance controls.
- Integrated SAST, SCA, and DAST security scanning into CI/CD pipelines for early vulnerability detection and remediation.
- Implemented DevSecOps governance controls to ensure compliance with enterprise security, audit, and regulatory requirements.
- Strengthened security posture through centralized vulnerability management and reporting mechanisms.
- Enhanced monitoring, logging, and alerting capabilities, enabling proactive incident detection and response.
- Improved system reliability by reducing change failure rates and Mean Time to Recovery (MTTR).
- Improved environment consistency, reducing configuration drift and deployment failures across environments.
- Standardized engineering practices and DevOps workflows across development, QA, security, and operations teams.
- Reduced manual effort and operational overhead through CI/CD and test automation.

Senior Software Engineer Jun 2020 – Jul 2021

Adanian Labs Africa | Nairobi, Kenya

- Automated infrastructure provisioning using Terraform and CloudFormation, guaranteeing consistent, zero-drift deployments and reducing manual setup time across all environments.
- Significantly improved code quality and reliability by implementing automated testing frameworks and enforcing their use across the development lifecycle.
- Expanded application capabilities and service reach by integrating critical third-party APIs, supporting new features for the user base.
- Accelerated product delivery by developing, testing, and maintaining core software applications using Java technologies.
- Enhanced software usability and user satisfaction by actively collaborating with UI/UX designers to integrate user-centric design improvements.
- Ensured efficient project delivery and stakeholder visibility by utilizing Agile methodologies to manage project timelines and deliverables.
- Reduced knowledge transfer friction and minimized future maintenance costs by thoroughly documenting software designs and architecture.
- Secured business continuity by devising and implementing automated backup and recovery protocols to ensure data preservation and integrity.
- Strengthened application security posture by conducting security assessments and implementing robust access controls, authentication, and encryption measures.
- Successfully launched the Ecobba startup product by designing and developing the complete Android application and backend services using Java, Kotlin, Node.js, Express.js, and MongoDB.
- Enabled key user functionality (transactions and logistics) by implementing user registration, secure payment integration, and real-time location tracking (Google Maps API and Firebase).
- Achieved high scalability and operational cost efficiency by deploying backend services on AWS using serverless architectures (AWS Lambda and API Gateway).

Senior Android Developer Feb 2019 – May 2020

Mobitech Technologies | Nairobi, Kenya

- Designed and implemented intuitive user interfaces ensuring a seamless user experience and adherence to Material Design guidelines.
- Integrated Android applications with backend services using RESTful APIs, handling data synchronization and real-time updates.
- Implemented data persistence using SQLite, ensuring efficient and reliable storage of application data.
- Conducted unit testing and integration testing to identify and fix issues, ensuring application stability and performance.
- Worked closely with backend developers to define API requirements and troubleshoot integration issues.
- Assisted in writing Python scripts for automating log file analysis and extracting key metrics for troubleshooting.
- Gained hands-on experience with Linux system administration tasks including file permissions, process management, and shell scripting.
- Utilized AWS services such as EC2, S3, and Lambda for cloud-based application development and deployment.

Software Engineer Jan 2018 – Jan 2019

EldoHub Innovation Center | Nairobi, Kenya

- Led the development of a cloud-based inventory management system utilizing AWS infrastructure and microservices architecture.
- Designed and implemented scalable and efficient backend systems using Node.js, Express.js, and MongoDB.
- Developed and maintained front-end systems using React and Material UI, ensuring a seamless user experience.
- Utilized Docker and Kubernetes for containerization and orchestration, ensuring high availability and scalability of the system.
- Conducted code reviews and implemented automated testing to ensure code quality and maintainability.

- Led a team of developers, providing technical guidance and support throughout the software development lifecycle.
- Utilized Jira for effective project management, ensuring timely delivery and effective communication with stakeholders.
- Applied security frameworks and standards such as OWASP and SonarQube throughout the software development lifecycle.

EDUCATION

Master of Science — Artificial IntelligenceIn Progress

The Open University of Kenya | Nairobi, Kenya

Dissertation (completed): “An AI-Based Predictive Model for Business Permit Revenue Forecasting in Nandi County, Kenya” developed and evaluated a machine-learning model forecasting county own-source (business permit) revenue from ward-level records, applying a Design Science Research methodology and benchmarking against traditional county budget-estimation approaches.

Bachelor of Science — Computer ScienceDecember 2019

Dedan Kimathi University of Technology | Nyeri, Kenya

Coursework: Algorithms and Data Structures, Operating Systems, Computer Networks, Database Systems, Object-Oriented Programming, Software Engineering, Cybersecurity, Cryptography, Network Security

CERTIFICATIONS

- CKS — Certified Kubernetes Security Specialist
- CKA — Certified Kubernetes Administrator
- CKAD — Certified Kubernetes Application Developer
- HashiCorp Certified Terraform Associate

REFERENCES

Nixon Kipkorir | +254 702 739 804 | Safaricom PLC | nixswinner@gmail.com

Susan Kwamboka | +254 705 137 818 | Mastercard Foundation | susankwamboka81@gmail.com

Stanley Mbote | +254 705 822 035 | Software Engineer, The Jitu | starnleymbote@gmail.com

